

Supplementary Material

Combining Mutual Information and Strangeness Minimization for Feature Selection in Conformal Prediction

Marcos López-De-Castro^{1,2,*}, Alberto García-Galindo^{1,2}, and Rubén Armañanzas^{1,2}

¹Institute of Data Science and Artificial Intelligence (DATAI), University of Navarra, Ismael Sánchez Bella Building, Campus Universitario, Pamplona, Navarre, Spain.

²TECNUN School of Engineering, University of Navarra, Donostia-San Sebastián, Basque Country, Spain.

*Corresponding: mlopezdecas@unav.es

1 Synthetic Datasets

For each synthetic network, valid target variables were first identified by requiring the target to have at least one parent, one child, and one spouse, so that its Markov blanket was non-trivial. Targets were further filtered to ensure suitability for supervised evaluation: discrete targets were required to satisfy a minimum class-balance threshold, while continuous targets were allowed only when they could be discretized into quantile-based bins satisfying a minimum bin-frequency constraint. For each selected target, samples were drawn from the corresponding Bayesian network (using `bnlearn::rbn`) at three sample sizes, 250, 1000, and 5000, with one replication per configuration. The raw sampled datasets were then converted into the format used by the feature-selection pipeline: non-numeric binary variables were encoded as -1 and 1; non-numeric multiclass target variables were label-encoded; constant or near-empty predictors were removed; and non-numeric multiclass predictors were transformed using one-hot encoding with the first level dropped. One-hot transformation information was stored, allowing dummy variables to be mapped back to their original variables during Markov blanket evaluation. Finally, after feature selection, those selected dummy variables were grouped back to their source variables when computing Markov blanket recovery metrics, so that performance was assessed at the original-variable level. Further details are in Supplementary Table S1.

Dataset	Nodes	Targets	Markov Blanket		Avg.	Child.	Spous.	OHE	OHE	Vars	Cols	Width
			Mean	Median								
arth150	107	10	7.100	7.000	3.100	2.500	4.300	106.000	0.000	0.000	0.000	0.000
ecoli70	46	10	6.700	6.500	1.800	3.400	3.100	45.000	0.000	0.000	0.000	0.000
healthcare	7	4	3.250	3.500	1.500	1.000	1.000	8.000	2.000	4.000	2.000	2.000
link	724	10	12.000	9.000	3.000	4.200	4.800	609.200	0.600	1.200	0.400	0.400
magic-irri	64	10	11.500	7.500	2.300	2.000	8.400	63.000	0.000	0.000	0.000	0.000
magic-niab	44	10	12.500	11.000	2.000	1.900	9.300	43.000	0.000	0.000	0.000	0.000
mehra-complete	24	10	16.800	16.500	5.800	2.400	13.700	63.200	4.000	44.200	28.200	28.200
pathfinder	109	4	11.000	12.500	1.250	9.000	4.250	284.750	67.500	254.250	47.250	47.250
sangiovese	15	10	12.100	12.000	3.800	3.800	10.300	28.000	1.000	15.000	15.000	15.000

Table S1: Dataset-level structural characteristics used in the synthetic benchmark. **Dataset** denotes the Bayesian network name; **Nodes** is the total number of variables in the original network; and **Targets** is the number of target variables evaluated for that dataset. The **Markov Blanket** columns report the mean and median Markov blanket size across the evaluated targets. **Avg. Par.**, **Avg. Child.**, and **Avg. Spous.** denote, respectively, the average number of parents, children, and spouses in the Markov blanket of the evaluated targets. **OHE Feat.** is the average number of features after one-hot encoding, **OHE Vars** is the average number of original variables expanded by one-hot encoding, **Dummy Cols** is the average number of dummy columns introduced, and **Max Width** is the average maximum number of dummy columns generated by any single encoded variable.

2 Additional Results

2.1 Synthetic Datasets

The supplementary material related to the results and discussion of the synthetic datasets is reported in the Supplementary Table S2.

Dataset	BTR	BTR(250)	BTR(1000)	BTR(5000)	Nodes	MB-mean	Spouses-mean	Features-OHE	Dummy-OHE	W/T vs SMFS	W/T vs mRMR
mehra-complete	0.833	1.000	1.000	0.500	24	16.800	13.700	63.200	44.200	0.833	1.000
pathfinder	0.750	0.750	1.000	0.500	109	11.000	4.250	284.750	254.250	1.000	0.750
link	0.706	0.333	1.000	0.800	724	12.000	4.800	609.200	1.200	1.000	0.706
ecoli70	0.700	0.600	0.700	0.800	46	6.700	3.100	45.000	0.000	0.833	0.867
magic-irri	0.533	0.300	0.600	0.700	64	11.500	8.400	63.000	0.000	0.667	0.767
healthcare	0.500	0.500	0.500	0.500	7	3.250	1.000	8.000	4.000	0.917	0.583
magic-miab	0.467	0.500	0.500	0.400	44	12.500	9.300	43.000	0.000	0.600	0.733
sangiovese	0.633	0.300	0.900	0.700	15	12.100	10.300	28.000	15.000	0.867	0.733

Table S2: Focused evidence table for the datasets discussed in the text, combining mRMR-MS best-or-tie rates, pairwise win-or-tie rates against the two components, and structural descriptors of each Bayesian network. BTR is teh Best-or-tie rate. Features-OHE refers to the number of features after one-hot-encoding. Dummy-OHE refers to the one-hot dummy cols, on average. W/T stands for Win or Tie.

2.2 Benchmark datasets

Next, we present the performance scores categorized by method, dataset, and the cardinality of selected features.

2.2.1 SPECTF

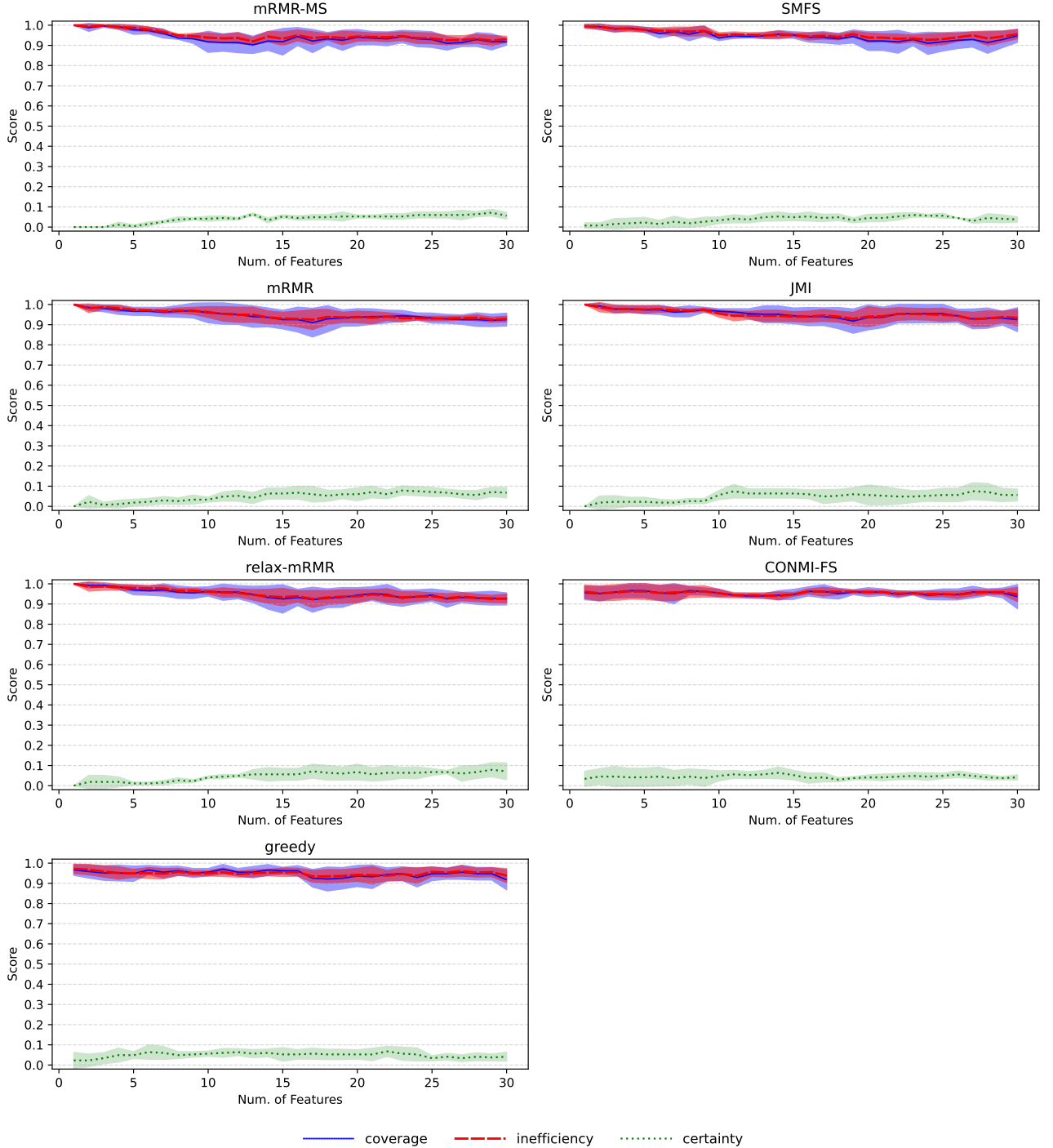


Figure S1: Performance metrics for the SPECTF dataset using a linear SVM classifier. Each panel corresponds to a feature-selection method and shows coverage, inefficiency, and certainty as a function of the number of selected features. Solid lines represent the mean score, and shaded bands indicate variability across repeated runs.

2.2.2 NAT Response

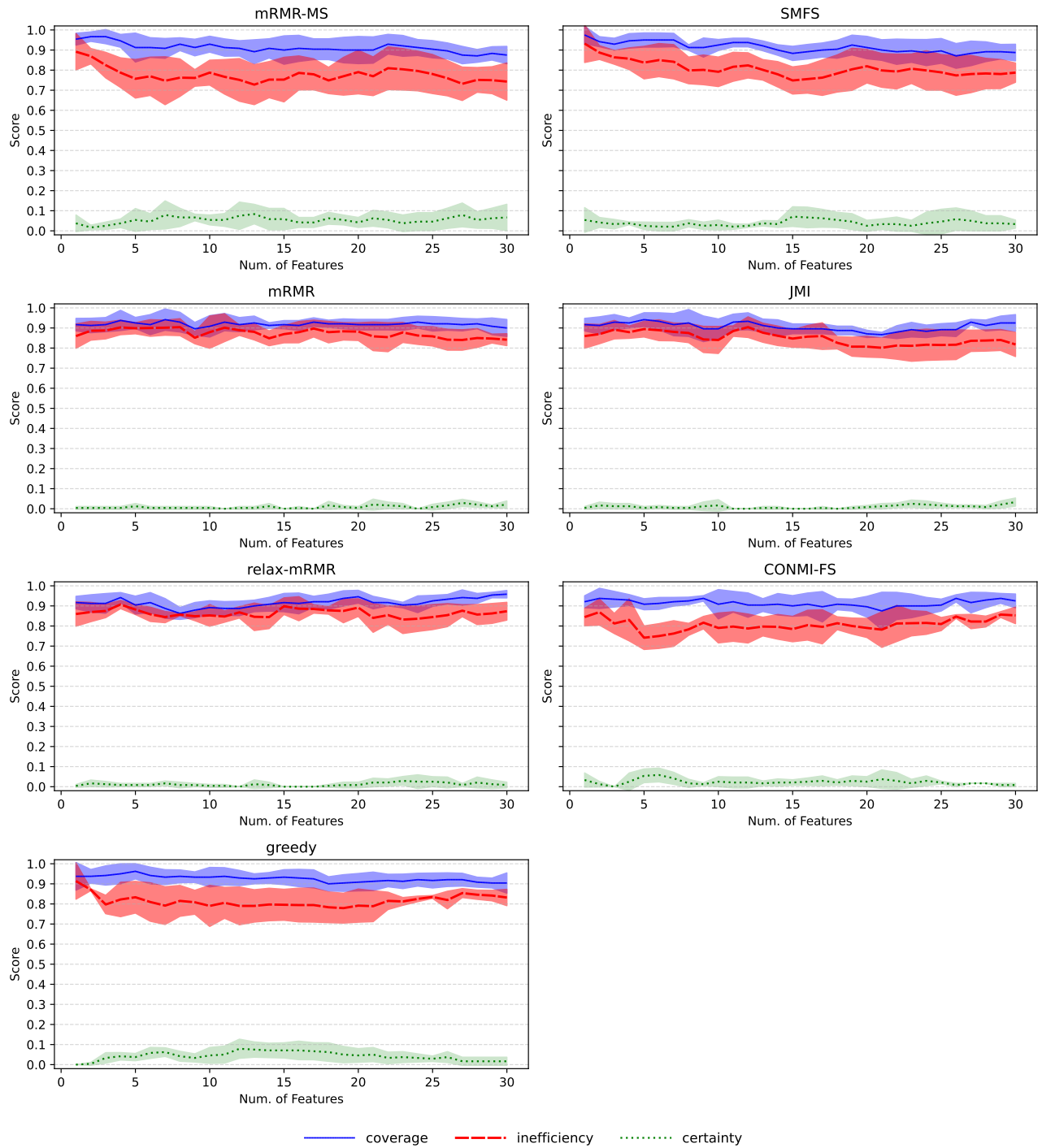


Figure S2: Performance metrics for the NAT Response dataset using a linear SVM classifier. Each panel corresponds to a feature-selection method and shows coverage, inefficiency, and certainty as a function of the number of selected features. Solid lines represent the mean score, and shaded bands indicate variability across repeated runs.

2.2.3 Coronary artery

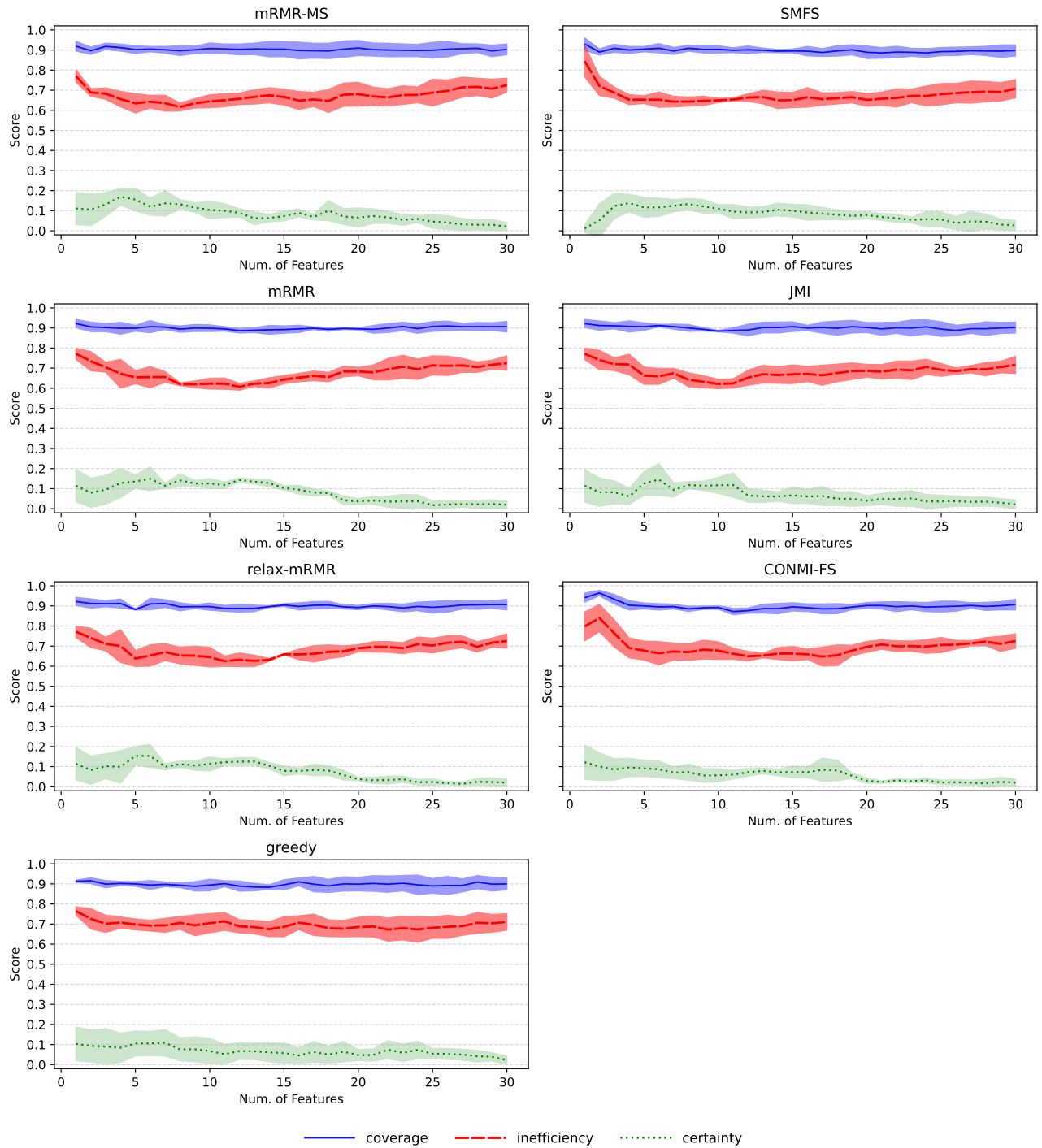


Figure S3: Performance metrics for the coronary artery dataset using a linear SVM classifier. Each panel corresponds to a feature-selection method and shows coverage, inefficiency, and certainty as a function of the number of selected features. Solid lines represent the mean score, and shaded bands indicate variability across repeated runs.

2.2.4 Forest type

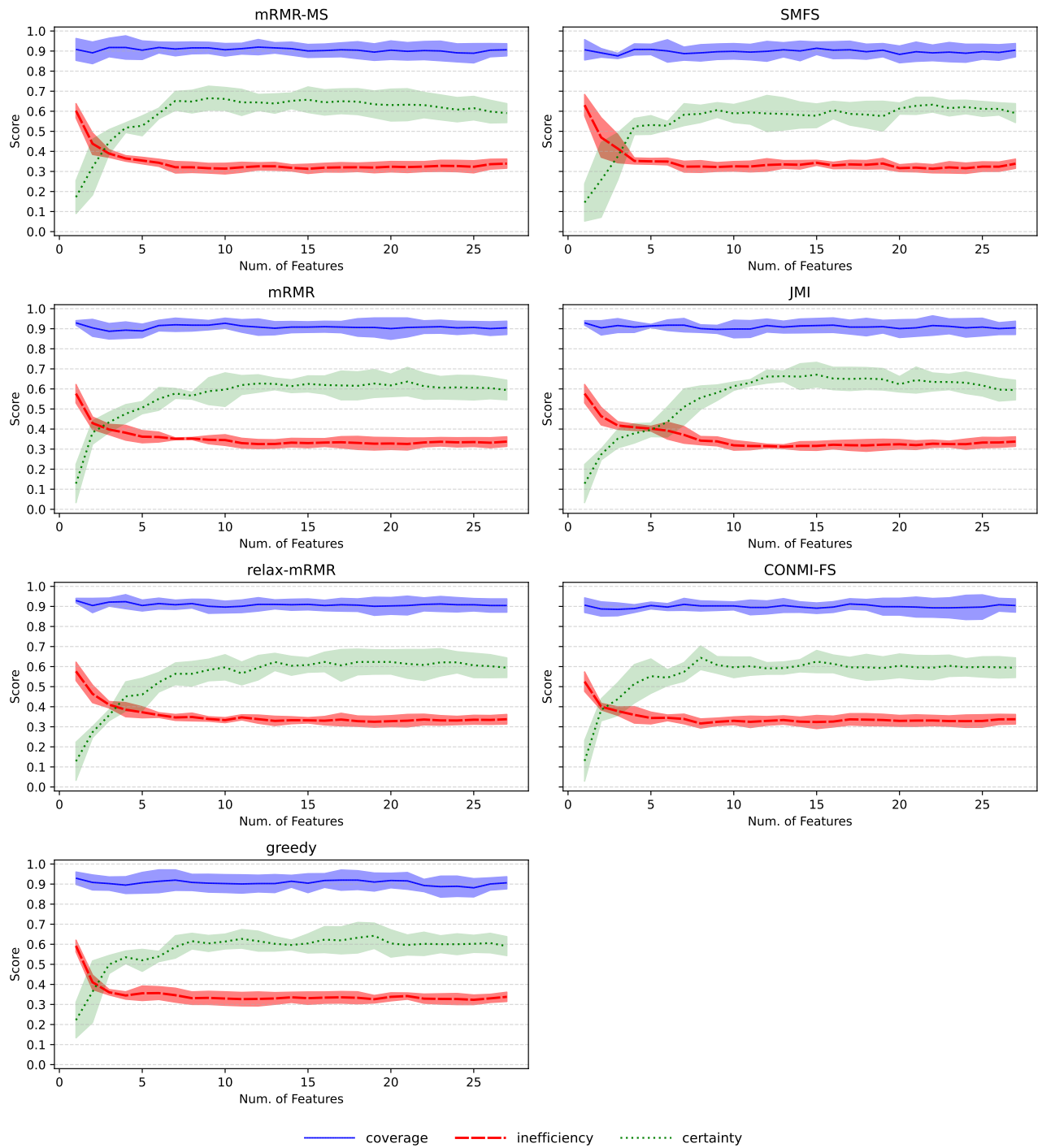


Figure S4: Performance metrics for the Forest Type dataset using a linear SVM classifier. Each panel corresponds to a feature-selection method and shows coverage, inefficiency, and certainty as a function of the number of selected features. Solid lines represent the mean score, and shaded bands indicate variability across repeated runs.

2.2.5 Ionosphere

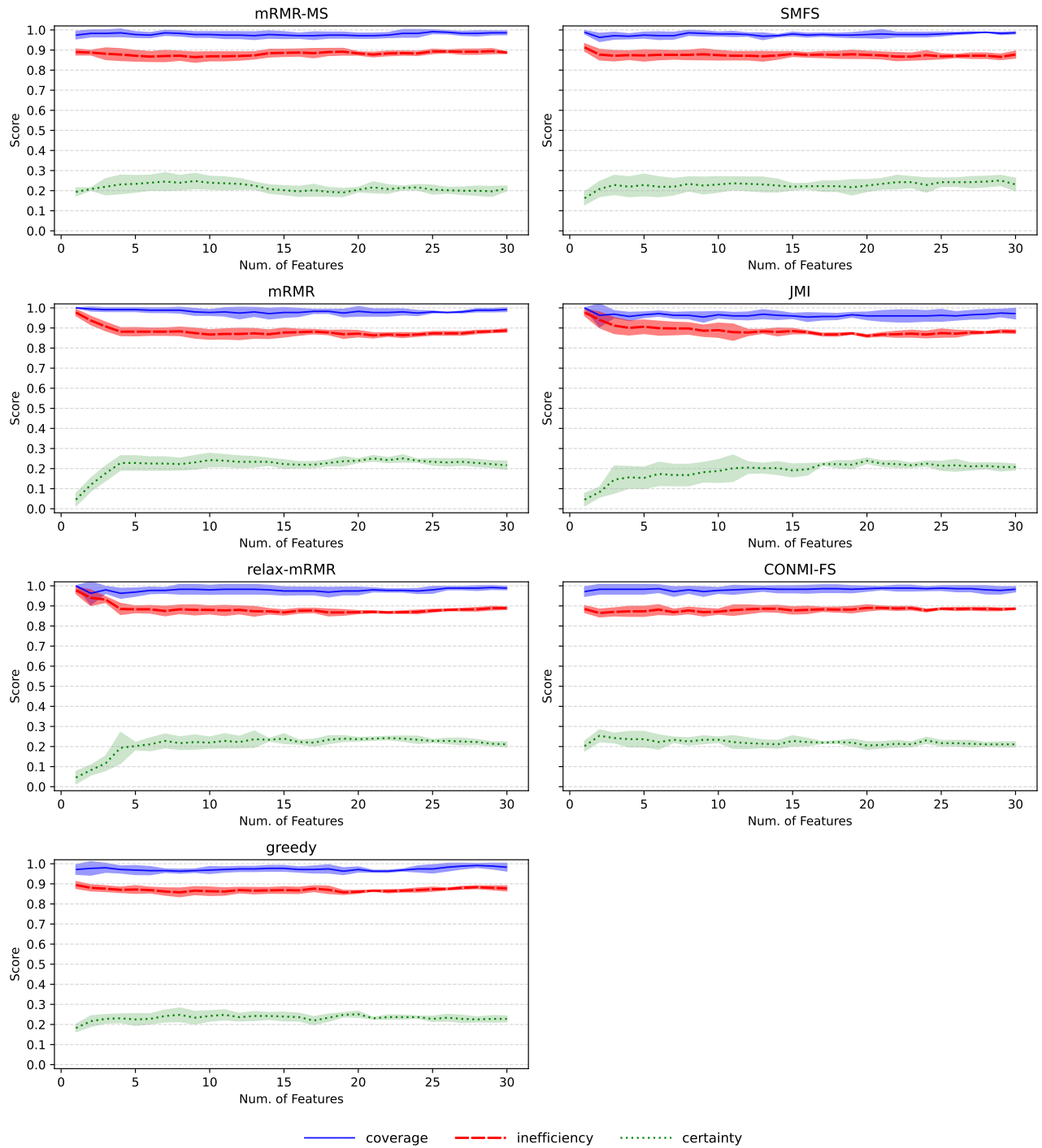


Figure S5: Performance metrics for the Ionosphere dataset using a linear SVM classifier. Each panel corresponds to a feature-selection method and shows coverage, inefficiency, and certainty as a function of the number of selected features. Solid lines represent the mean score, and shaded bands indicate variability across repeated runs.

2.2.6 Nb-1

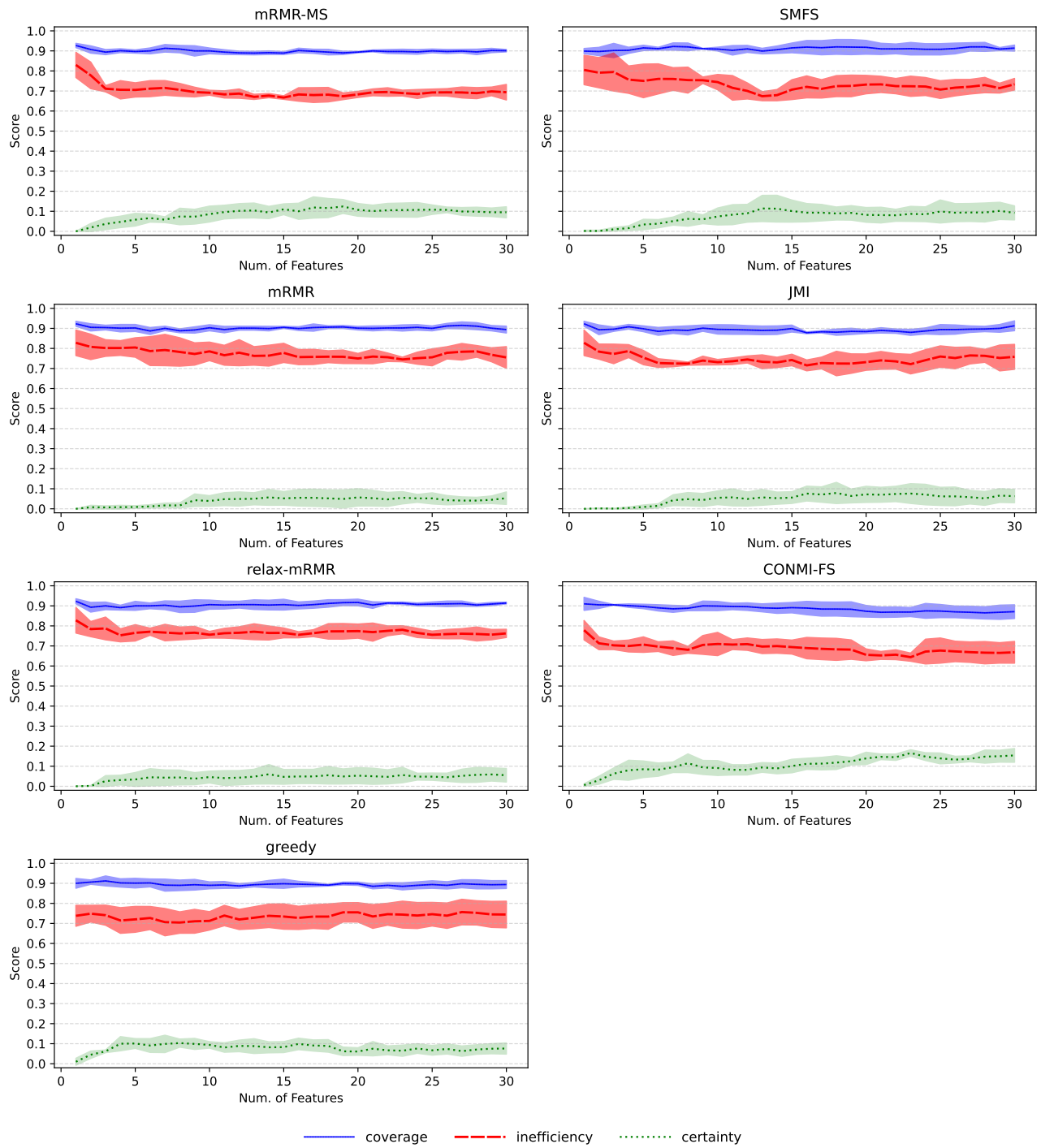


Figure S6: Performance metrics for the Nb-1 dataset using a linear SVM classifier. Each panel corresponds to a feature-selection method and shows coverage, inefficiency, and certainty as a function of the number of selected features. Solid lines represent the mean score, and shaded bands indicate variability across repeated runs.

2.2.7 Musk v1

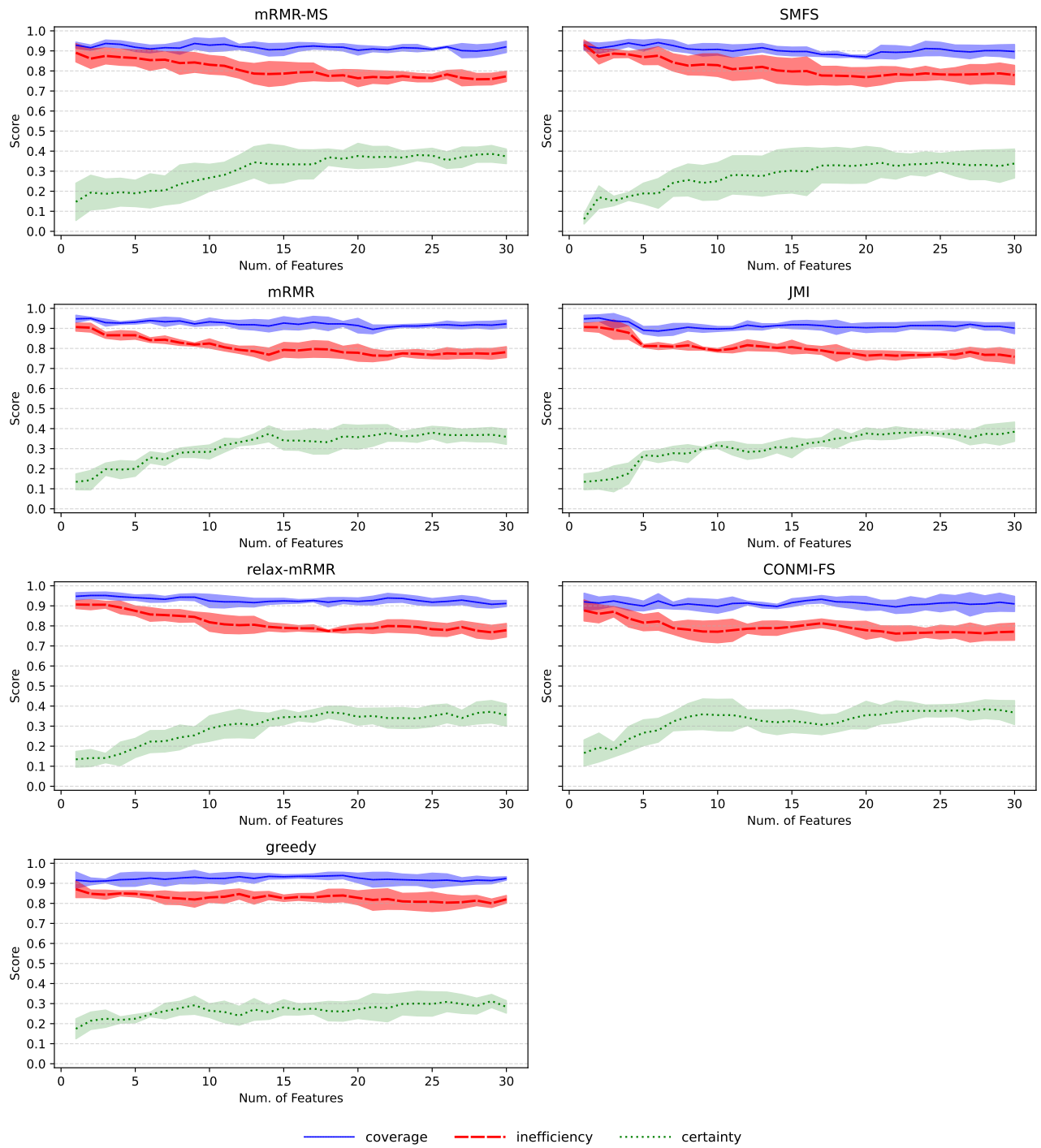


Figure S7: Performance metrics for the Musk v1 dataset using a linear SVM classifier. Each panel corresponds to a feature-selection method and shows coverage, inefficiency, and certainty as a function of the number of selected features. Solid lines represent the mean score, and shaded bands indicate variability across repeated runs.

2.2.8 Optdigits

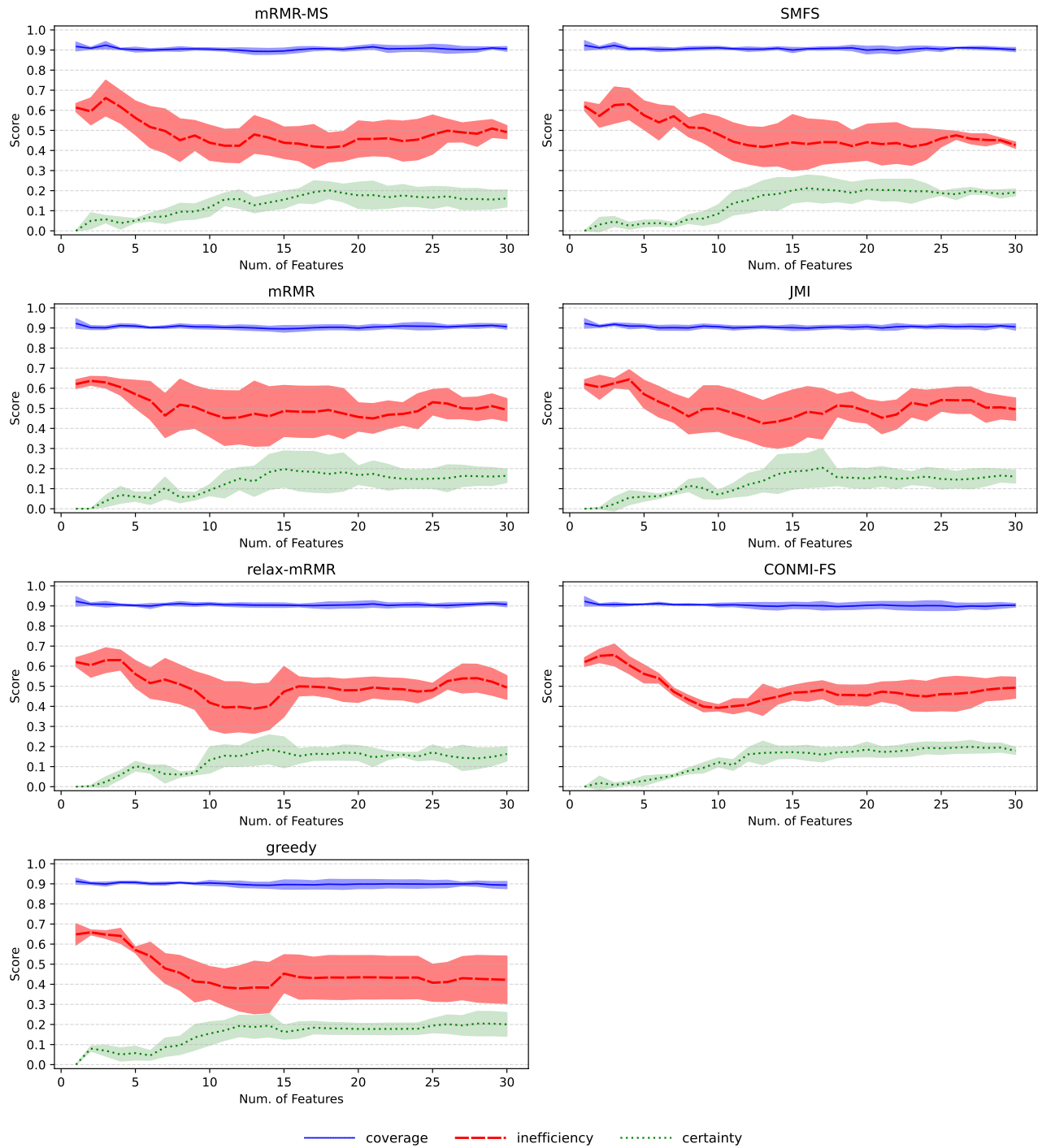


Figure S8: Performance metrics for the Optdigits dataset using a linear SVM classifier. Each panel corresponds to a feature-selection method and shows coverage, inefficiency, and certainty as a function of the number of selected features. Solid lines represent the mean score, and shaded bands indicate variability across repeated runs.

2.2.9 Sonar

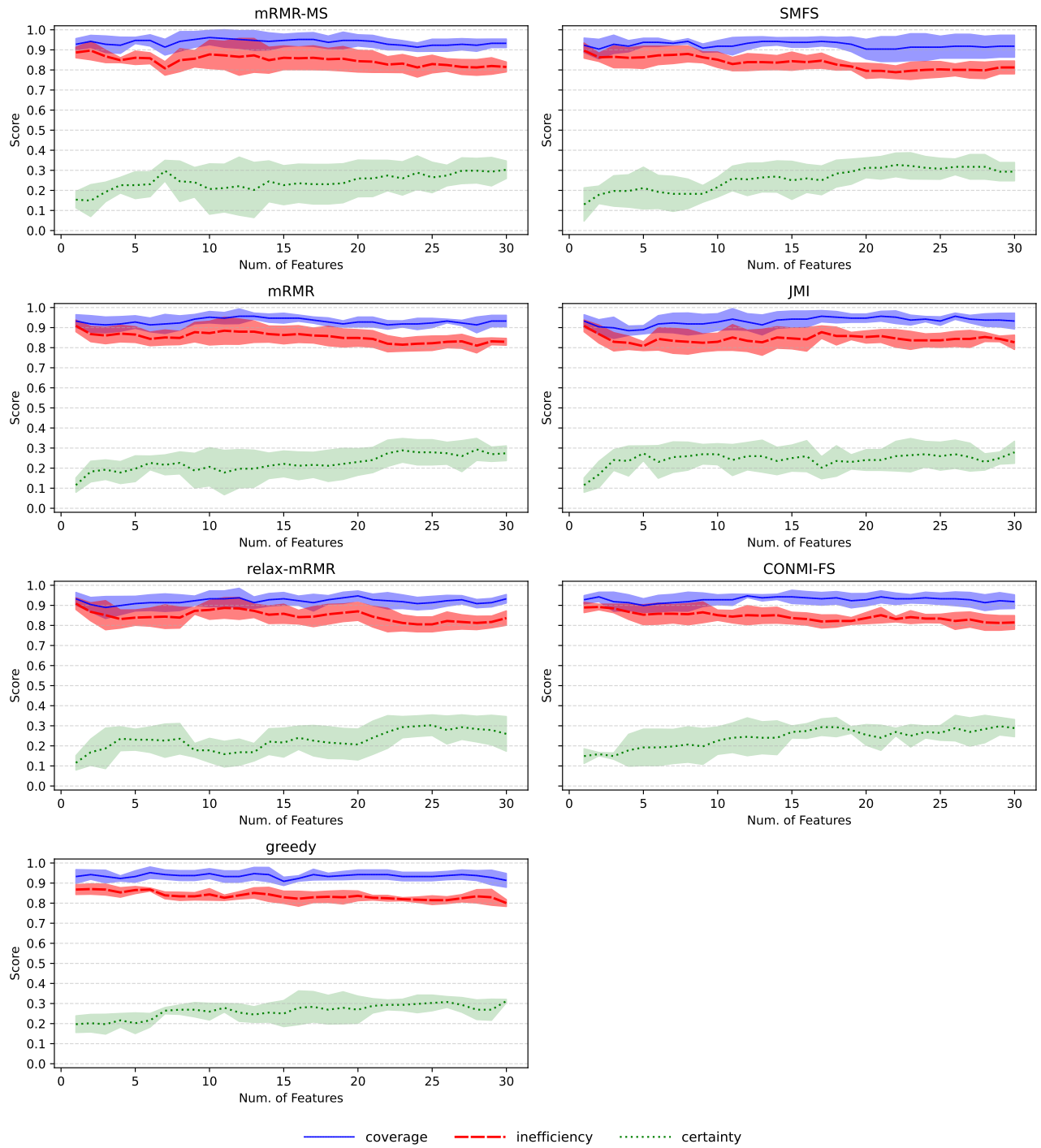


Figure S9: Performance metrics for the Sonar dataset using a linear SVM classifier. Each panel corresponds to a feature-selection method and shows coverage, inefficiency, and certainty as a function of the number of selected features. Solid lines represent the mean score, and shaded bands indicate variability across repeated runs.

2.2.10 Spambase

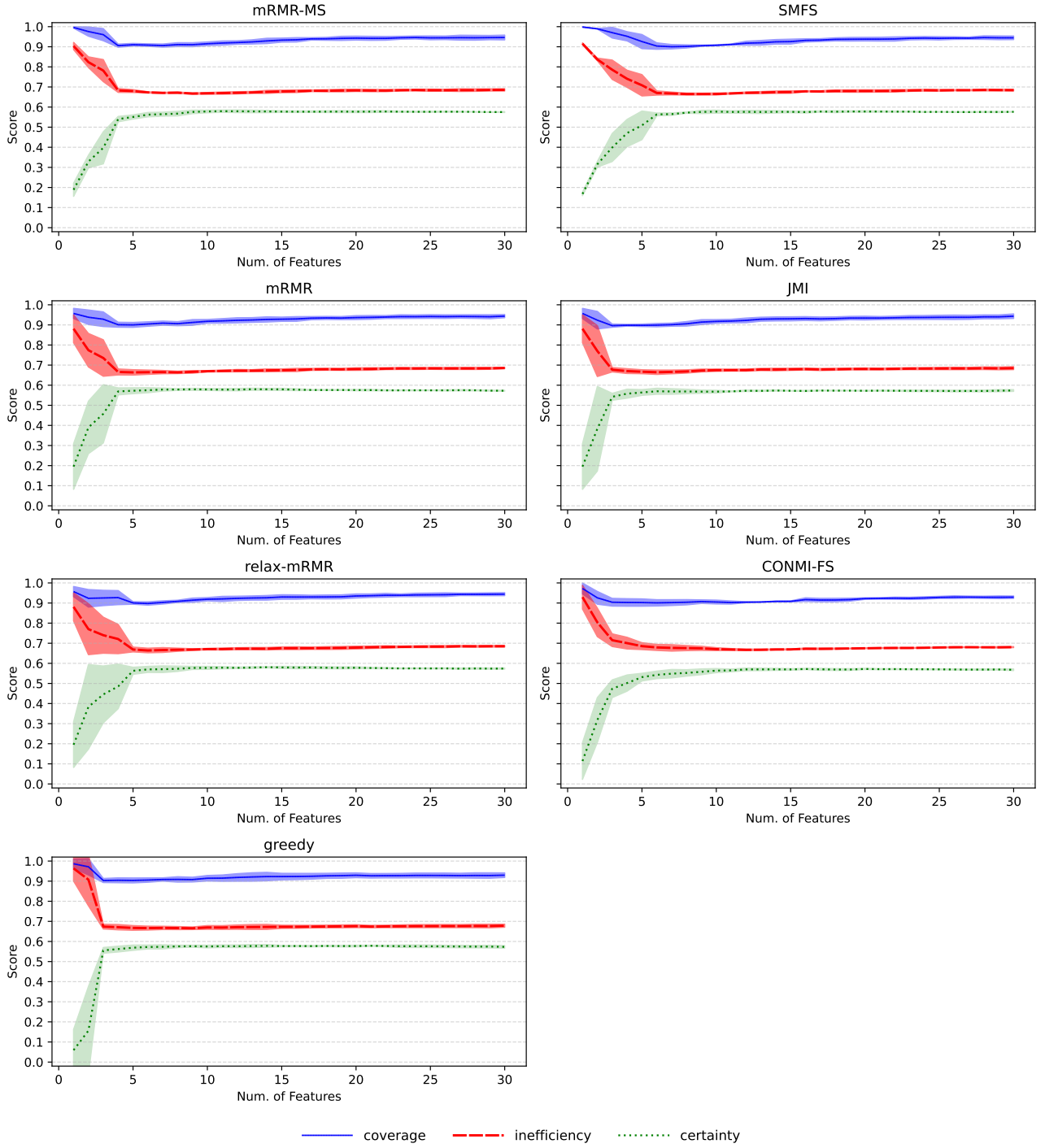


Figure S10: Performance metrics for the Spambase dataset using a linear SVM classifier. Each panel corresponds to a feature-selection method and shows coverage, inefficiency, and certainty as a function of the number of selected features. Solid lines represent the mean score, and shaded bands indicate variability across repeated runs.

3 Runtime comparison

Figure S11 shows the runtime performance comparison between the methodologies described in the main manuscript. To ensure a fair and controlled comparison, the experiments were performed on the same machine and with consistent random seeds. The analysis was performed over synthetic datasets designed with a varying number of classes, features, and samples, generated through *make_classification* function provided by

the *scikit-learn* v1.7. The reported time for each feature selection method corresponds to the computational time spent on selecting 10 features.

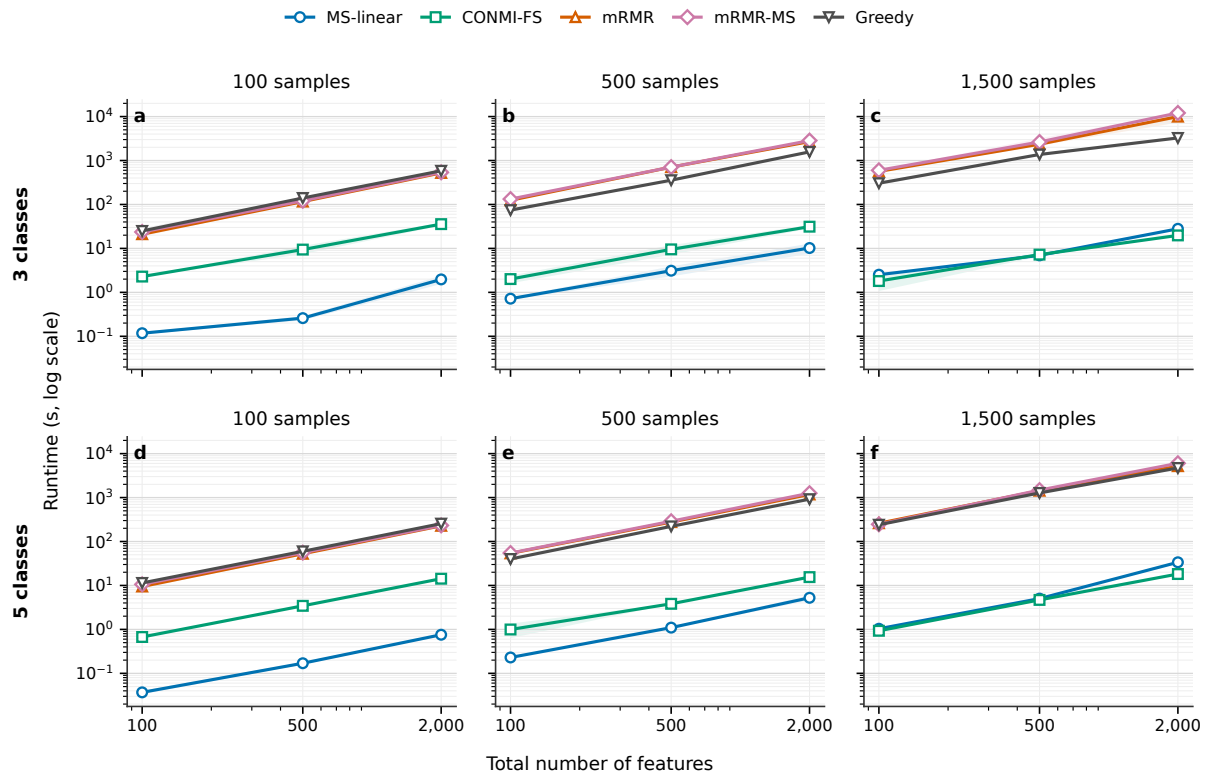


Figure S11: Runtime versus feature dimensionality. Panels show all combinations of the number of classes (3, 5) and sample sizes (500, 1500). The X-axis shows the total number of features of the dataset, whereas the Y-axis maps the running time (logarithmic scale) needed by each algorithm to select the first 10 relevant features.